

AI for Data Visualization Assignment

Q1. What is AI in Data Visualization? Explain how it differs from traditional data analysis.

Answer:

AI in Data Visualization uses artificial intelligence techniques such as machine learning and natural language processing to automatically analyze data, detect patterns, generate insights, and recommend visualizations. Traditional data analysis mainly depends on manual exploration and analyst expertise, whereas AI automates discovery of trends, anomalies, and predictions, making analysis faster and more accessible.

Q2. Explain how AI helps in automatic insight generation in BI tools.

Answer:

AI automatically scans datasets to identify trends, anomalies, correlations, outliers, and key drivers without requiring users to manually explore every chart. BI tools present these findings as suggested insights, helping users make quicker decisions.

Q3. What are the key AI features available in Microsoft Power BI? Explain any two with examples.

Answer:

Power BI AI features include Key Influencers, Smart Narratives, Anomaly Detection, Decomposition Tree, Q&A, and AutoML. Example features: Key Influencers identifies factors affecting sales; Smart Narratives automatically generates text summaries explaining dashboard results.

Q4. Explain the concept of Key Influencers and how AI identifies factors affecting a metric.

Answer:

Key Influencers is an AI visual in Power BI that identifies variables most strongly associated with a chosen outcome using statistical and machine learning techniques. For example, it may show that customer satisfaction increases when delivery time is under two days.

Q5. What is the role of Smart Narratives in AI-powered dashboards?

Answer:

Smart Narratives automatically create written summaries of charts and dashboards, highlighting

trends, increases, decreases, and important metrics, making reports easier to understand

Q6. Explain how Tableau Pulse provides proactive insights in Tableau.

Answer:

Tableau Pulse proactively monitors metrics and automatically notifies users about significant changes, trends, and explanations so they don't have to constantly check dashboards.

Q7. What is meant by AI-driven decision-making? How does AI support business decisions?

Answer:

AI-driven decision-making means using AI to analyze data and recommend actions. AI supports business decisions through forecasting, pattern detection, predictive analytics, automation, and personalized recommendations.

Q8. Explain the importance of interpreting AI-generated insights instead of directly trusting them.

Answer:

AI-generated insights should be interpreted critically because AI models can be affected by poor-quality data, bias, missing context, or incorrect assumptions. Human validation ensures accurate business decisions.

Q9. Discuss the limitations of AI in data visualization tools.

Answer:

Limitations include dependence on data quality, possible bias, lack of business context, limited explainability, privacy concerns, computational cost, and the need for human oversight.

Q10. Compare how AI is used differently in Power BI and Tableau.

Answer:

Power BI emphasizes built-in AI visuals, Azure AI integration, Copilot, AutoML, and natural language Q&A.; Tableau focuses on Explain Data, Tableau Pulse, Ask Data (legacy), Einstein AI integration, and intuitive visual analytics. Power BI is more tightly integrated with Microsoft's AI ecosystem, while Tableau emphasizes guided insights and visualization.